

ECS7040P: Statistical Foundations for Artificial Intelligence

Probability: Extra problems

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1. A bowl has 3 red and 2 yellow balls. Out of the bowl, I *randomly* select a ball.

Determine the following

Experiment:

Outcomes:

Sample Space:

Events:

Probability that I pick a red ball:



2. Let A, B, C be events from a random experiment I conduct. The probabilities are such that

$$P(A) = 0.45$$

$$P(B) = 1.2$$

$$P(C) = 20\%$$

What does $P(A)$ mean?

There is an error in the above probabilities. What is it?

3. Mark with a cross on the probability scale of *how likely* you think each event is.

(a) Getting a head on a coin toss.



(b) The sun rises tomorrow.



(c) You not attending today's class.





4. Two fair coins are tossed and the outcome is recorded. These are the events of interest

A : Observe at least one head

B : Observe at least one tail

Find the probabilities of events A, B, $A \cap B$, $A \cup B$ and A' .





5. Two cards are drawn from a deck of 52 cards. Calculate the probability that the draw includes an ace and a ten.



6. A sample is selected from either one of the two populations P_1 and P_2 , with probabilities of sample coming from each population being

$$P(P_1) = 0.7 \text{ and } P(P_2) = 0.3.$$

□ If the sample has been selected from P_1 , the probability of observing an event A is

$$P(A|P_1) = 0.2.$$

□ If the sample has been selected from P_2 , the probability of observing an event A is

$$P(A|P_2) = 0.3.$$

(a) If a sample is randomly selected, what is the probability that event A will occur?

(b) If a sample is randomly selected and event A is observed, what is the probability that sample is from P_1 ? From P_2 ?



